

Vipransh Ojha

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Education

VIT Bhopal University

B.Tech in Computer Science and Engineering (AI & ML Specialization)

Bhopal, India

Sep 2023 – May 2027 (Expected)

Technical Skills

- **Languages:** Python, TypeScript, JavaScript, C++, Java, SQL, HTML5, CSS3
- **Web Development & Backend:** React, Node.js, Express.js, Flask, FastAPI, REST APIs, Tailwind CSS
- **Databases Cloud:** MongoDB, PostgreSQL, SQLite, AWS Cloud Services, Docker, Git, Linux
- **AI/ML Data Science:** TensorFlow, PyTorch, Scikit-learn, OpenCV, Keras, Pandas, NumPy, NLP
- **Developer Tools Libraries:** OpenAI APIs, Google TTS, Whisper, MediaPipe, Pygame, Matplotlib

Experience

Software Development Intern

BISAG-N (Government of India)

New Delhi, India

May 2025 – Jun 2025

- Deployed a full-stack location recommendation system for 500+ users with 85% recommendation accuracy.
- Implemented content-based filtering and collaborative filtering algorithms using Python and Scikit-learn.
- Architected a scalable Flask backend with PostgreSQL, processing over 100 queries per second.
- Designed a responsive user interface with dynamic mapping which improved user engagement by 25%.

Research Intern (SERB Sponsored Project)

VIT Bhopal University

Bhopal, India

May 2024 – Jun 2024

- Built 'MolSpectra', a molecular visualization tool from scratch for a computational chemistry research project (SERB Grant No. CRG/20221002761).
- Engineered a unified interface to integrate 5+ quantum chemistry packages, including Gaussian and GAMESS, automating simulation workflows for researchers.
- Created the application's GUI using PyQt and implemented 3D molecular rendering with GLEW, streamlining the workflow from structure creation to calculation.

Projects

Forensic Identification using Dental Radiology

AI/ML Research Project

Jul 2025 – Present

Python, Siamese Neural Networks, TensorFlow, Keras

- Constructed an automated forensic identification system with 92% accuracy in matching dental records.
- Trained a Siamese Neural Network on the DENTEX dataset to compare radiographs.
- Crafted a Flask-based web interface for experts to upload and analyze AM/PM records efficiently.
- Optimized the system to process over 1,000 records per hour, reducing identification time by 90%.

SpeechSync - Voice-to-Text Synchronization System

Collaborative AI Project

Nov 2024 – Dec 2024

Python, NLP, Speech Processing

- Formulated a real-time voice-to-text synchronization system with 95% transcript alignment accuracy.
- Executed advanced speech processing pipelines using Python and NLP, handling 10+ hours of audio data.
- Fabricated an intuitive and user-friendly interface that improved transcript review speed by 30%.
- Revamped overall system performance for faster processing, reducing synchronization delays by 40%.

Achievements & Activities

- Solved over 100+ algorithmic problems on LeetCode.
- Competed in multiple coding competitions and hackathons.